IBM EMPLOYEE ATTRITION ANALYSIS

Machine Learning-based PoC

AGENDA / OVERVIEW

- Background & industry context
- Data & problem definition
- Exploratory Data Analysis (EDA)
- Machine learning model & results
- Business recommendations & impact
- Next steps

COMPANY CONTEXT – IBM

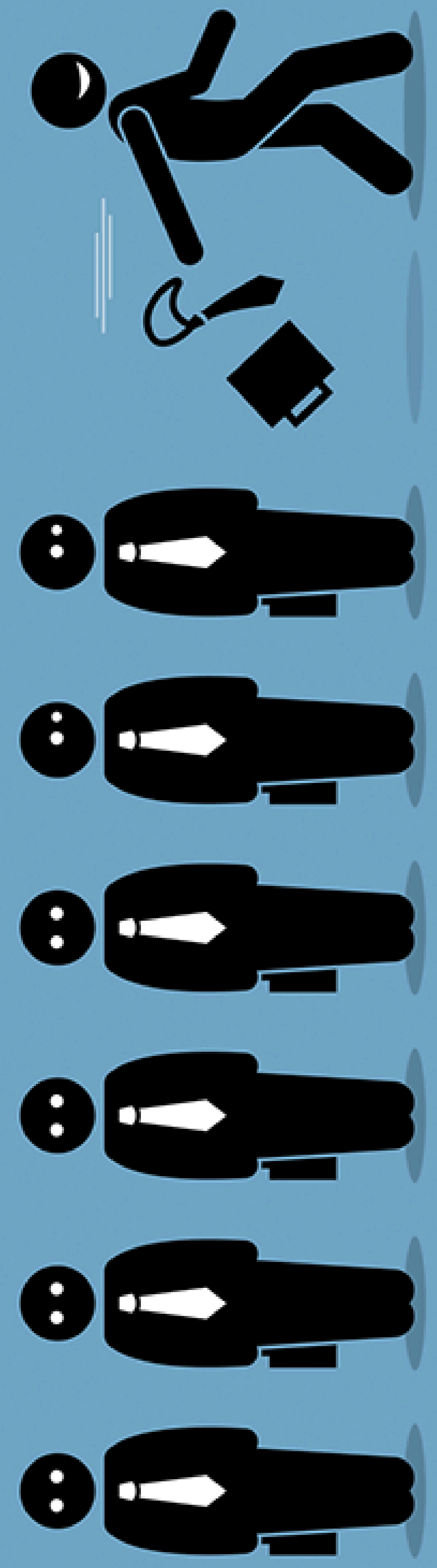
IBM is an IT consulting firm whose business heavily depends on retaining skilled tech and consulting talent, but it is facing challenges with employee retention.

- IBM’s consulting division = project-based, knowledge-intensive business.
- Losing employees → lost expertise, delivery risk, and higher costs.
- Our PoC uses IBM’s historical HR data with attrition labels to:
  - Understand who is at risk of leaving
  - Build a predictive model to support proactive retention.

INDUSTRY ATTRITION CONTEXT

IBM’s retention challenge is part of a broader industry problem where tech/IT firms structurally face high and costly attrition.

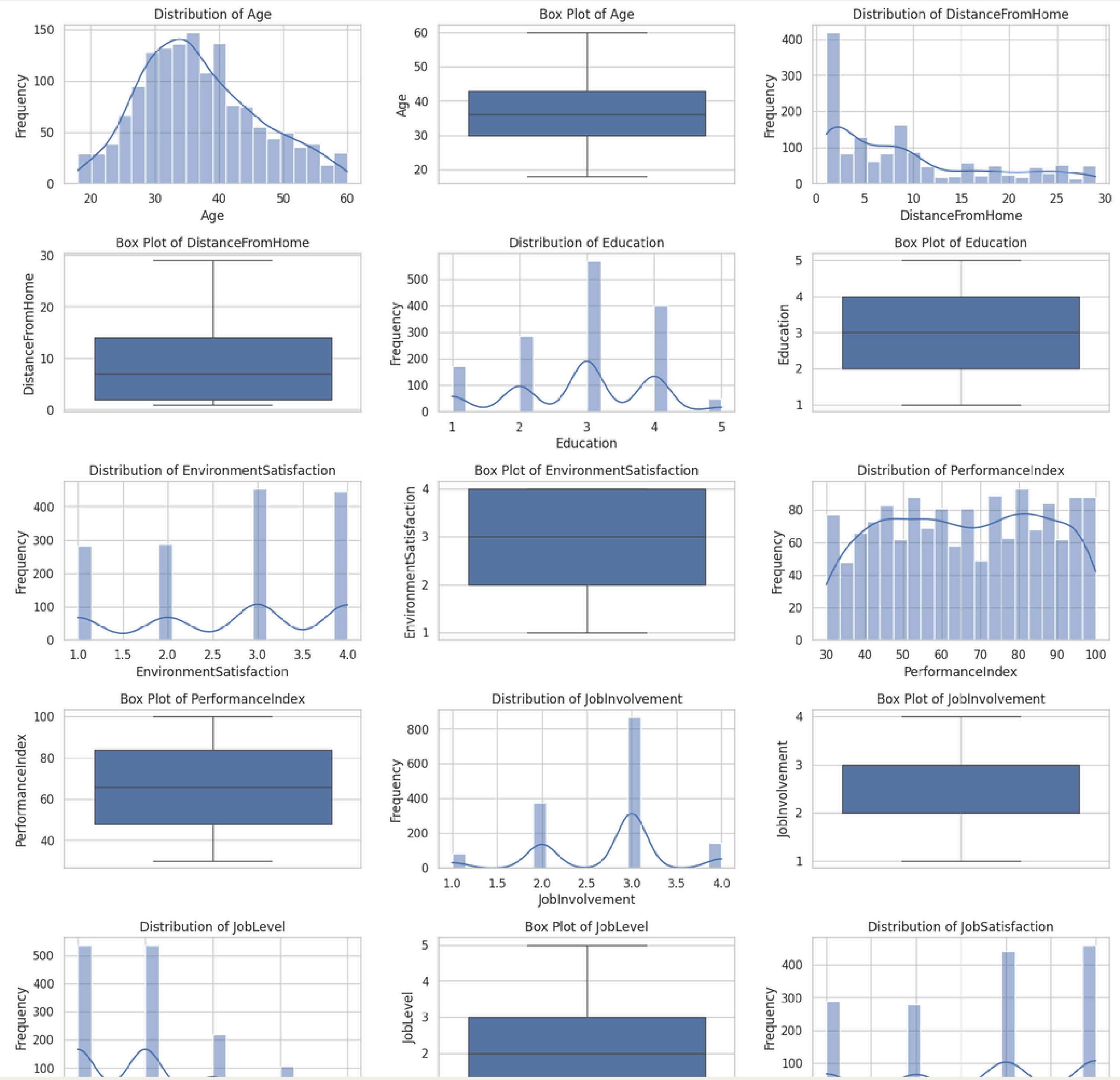
- IT firms aim for ~10% or less attrition, but real tech attrition is often much higher (around low-teens to 20%+).
- Turnover cost =
  - Direct: hiring, onboarding, training, ramp-up time
  - Indirect: lost productivity, knowledge loss, burnout, weaker culture, client risk
- Research: tech employees often leave due to:
  - Job role & growth dissatisfaction
  - Pay & incentives
  - Burnout & stress
  - Lack of recognition / unfair treatment
- These causes map exactly to IBM’s data: stress, incentives, remote/flex work, satisfaction, welfare benefits.



DATA & PROBLEM DEFINITION

We formulate attrition as a binary classification problem on IBM’s HR data.

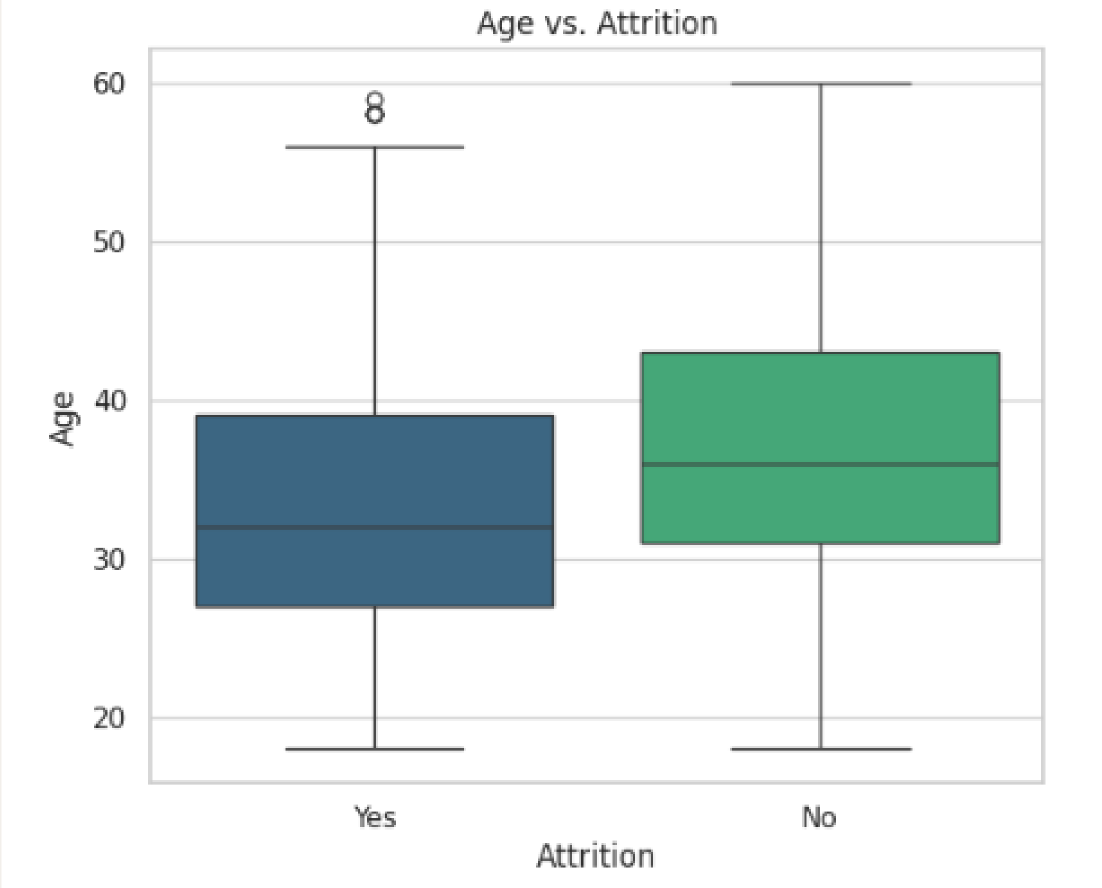
- Data: ~1,500 employees, 40+ HR features (demographics, job, stress, compensation, benefits, work style).
- Target variable: Attrition (Yes / No).
- ML Task:
    - Predict probability that an employee will leave.
    - Enable earlier, targeted interventions by HR.
- Key assumptions:
    - Data is representative of IBM’s consulting workforce in recent years.
    - Features like StressRating, RemoteWork, Incentive, WelfareBenefits are actionable levers.

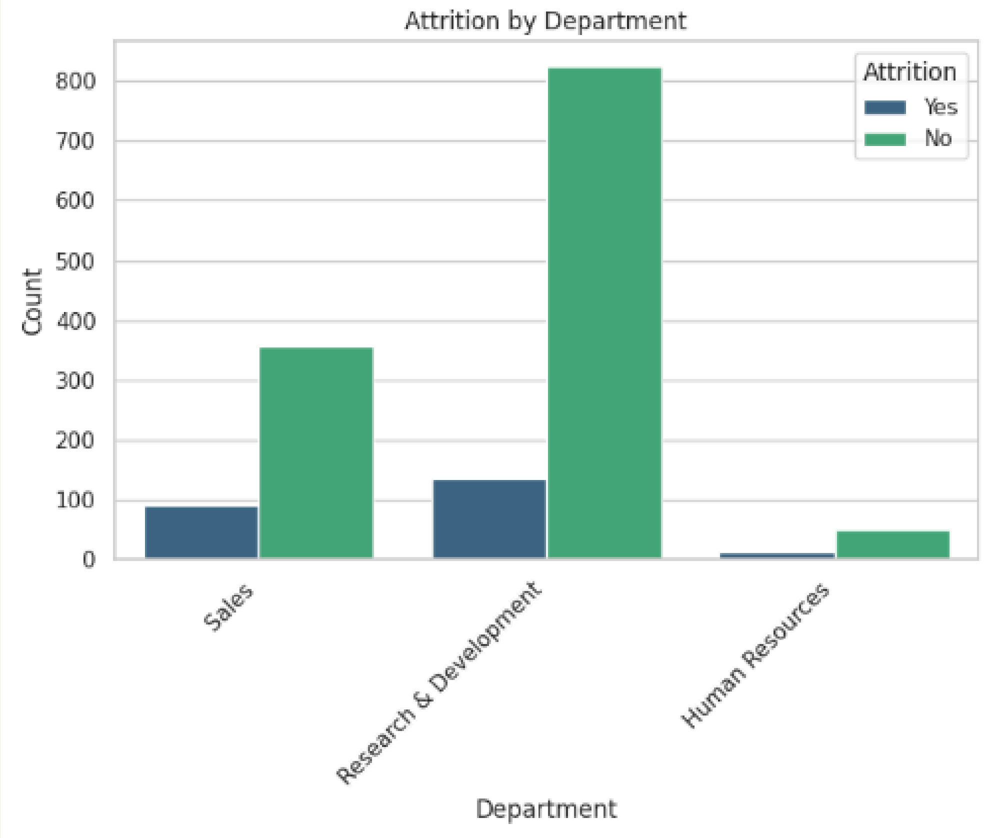


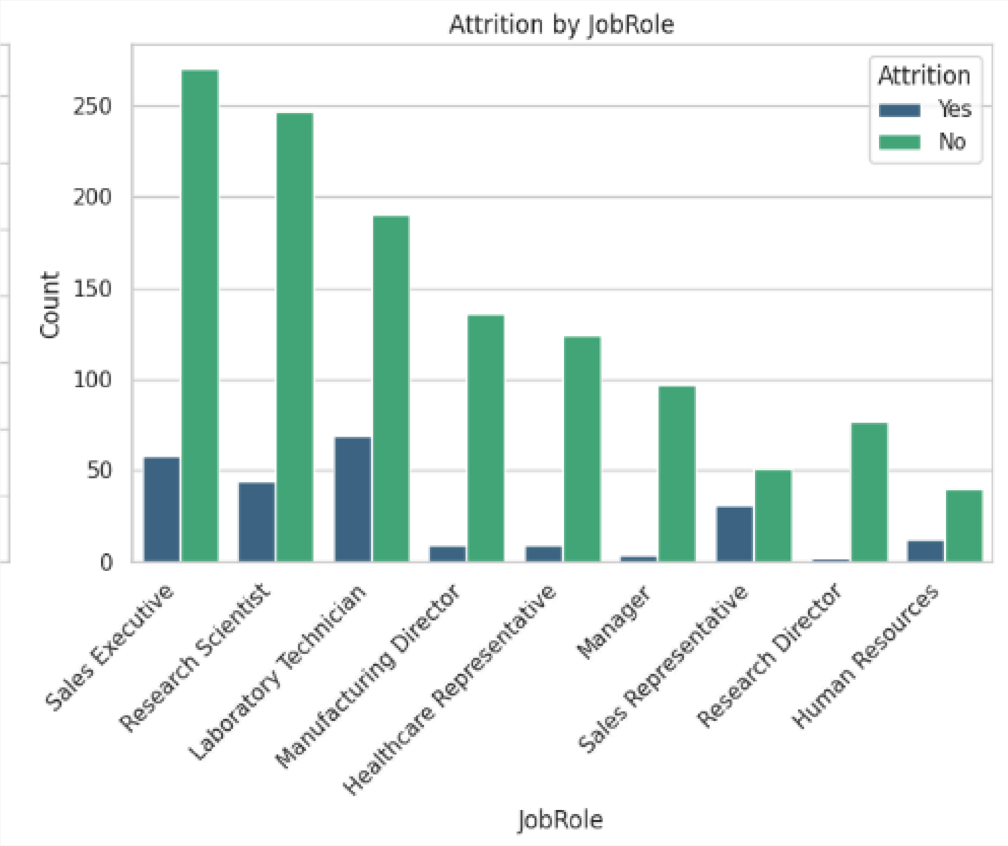
KEY EDA – WHO IS LEAVING?

Attrition is concentrated among younger, early-tenure employees and specific roles/departments.

- Younger employees and those with shorter tenure are more likely to leave.
- Higher attrition in certain roles (e.g., sales-related / frontline job roles) and specific departments.
- Singles / early career employees show higher risk compared to more established employees.



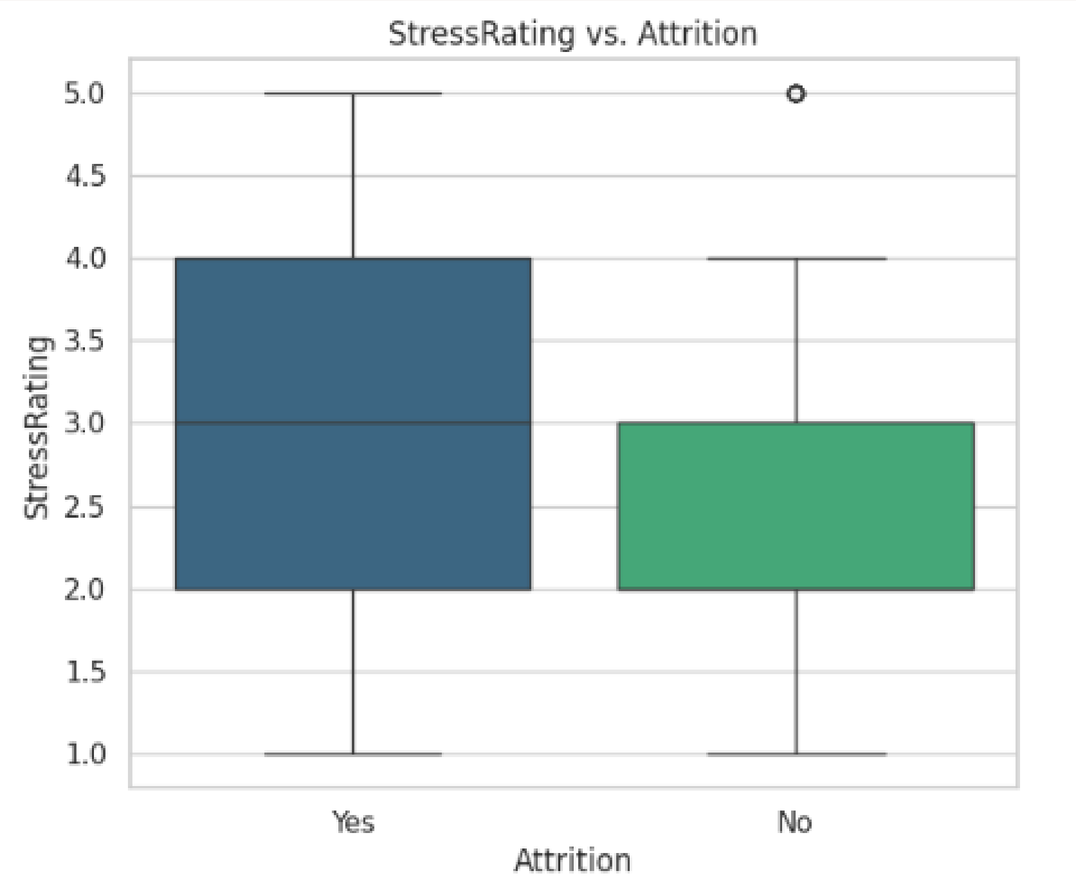


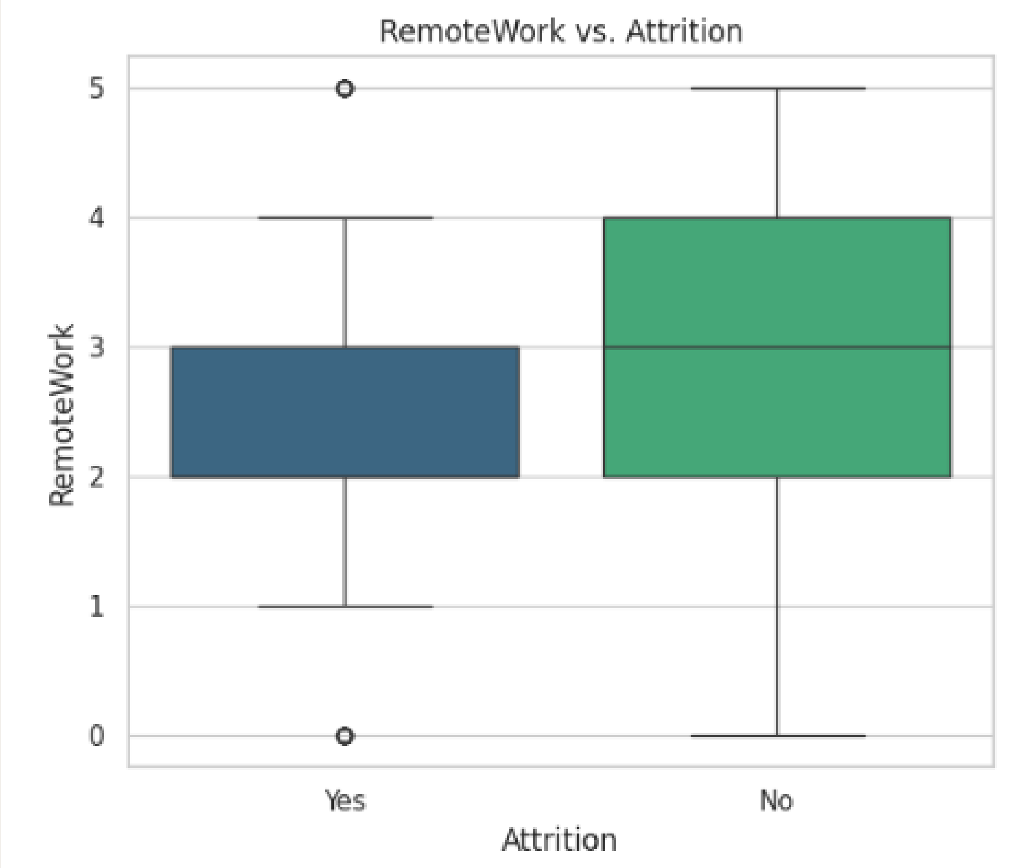


KEY EDA – WHY ARE THEY LEAVING? (WORK & STRESS)

High stress, low flexibility, and low incentive/benefit usage are strongly associated with attrition.

- StressRating (HR-assessed): higher stress → much higher likelihood of leaving.
- RemoteWork & FlexibleWork: on-site / no-flex employees are more likely to leave than those with remote/flexible arrangements.
- Incentive & MonthlyIncome: leavers tend to have lower incentives and pay.
- WelfareBenefits & facilities / extended leave: employees who actively use these tend to stay longer.





MODELLING APPROACH (ML POC)

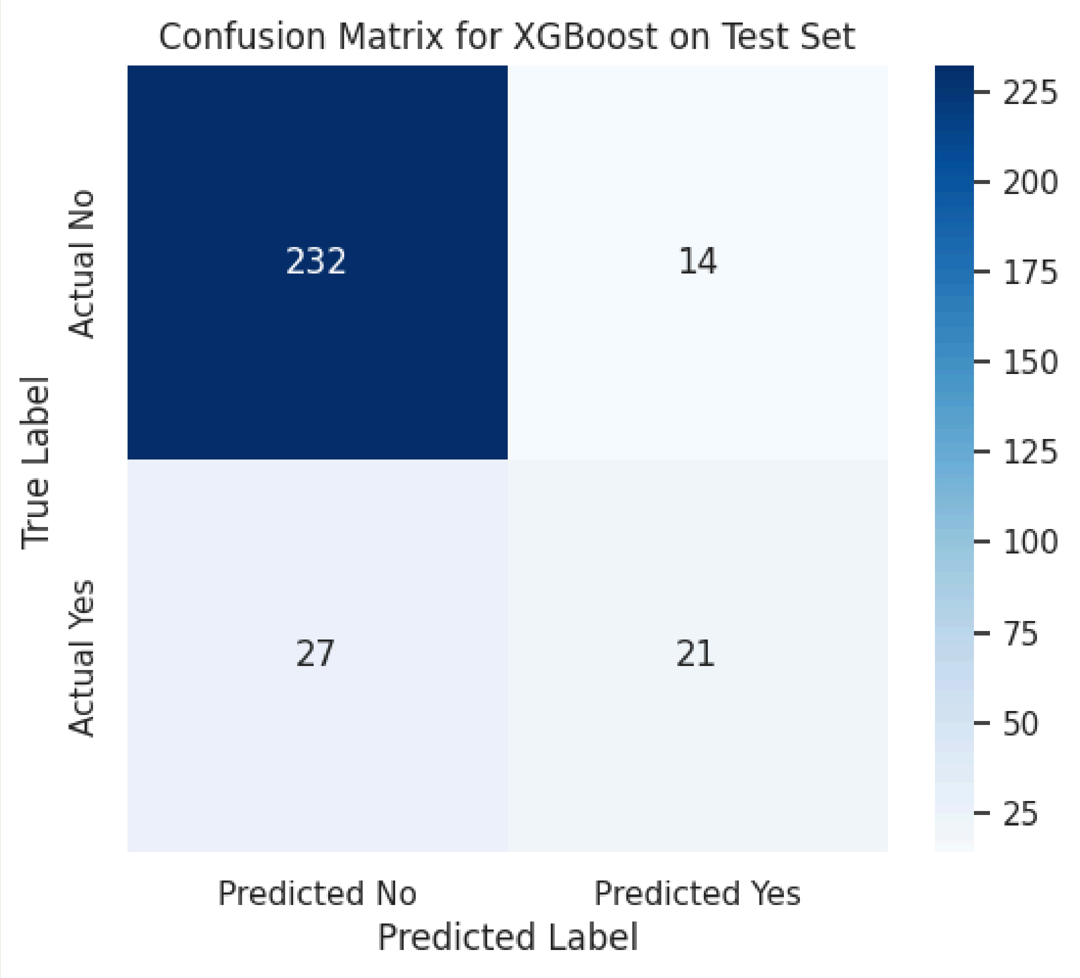
We build a straightforward, interpretable PoC using a modern tree-based classifier.

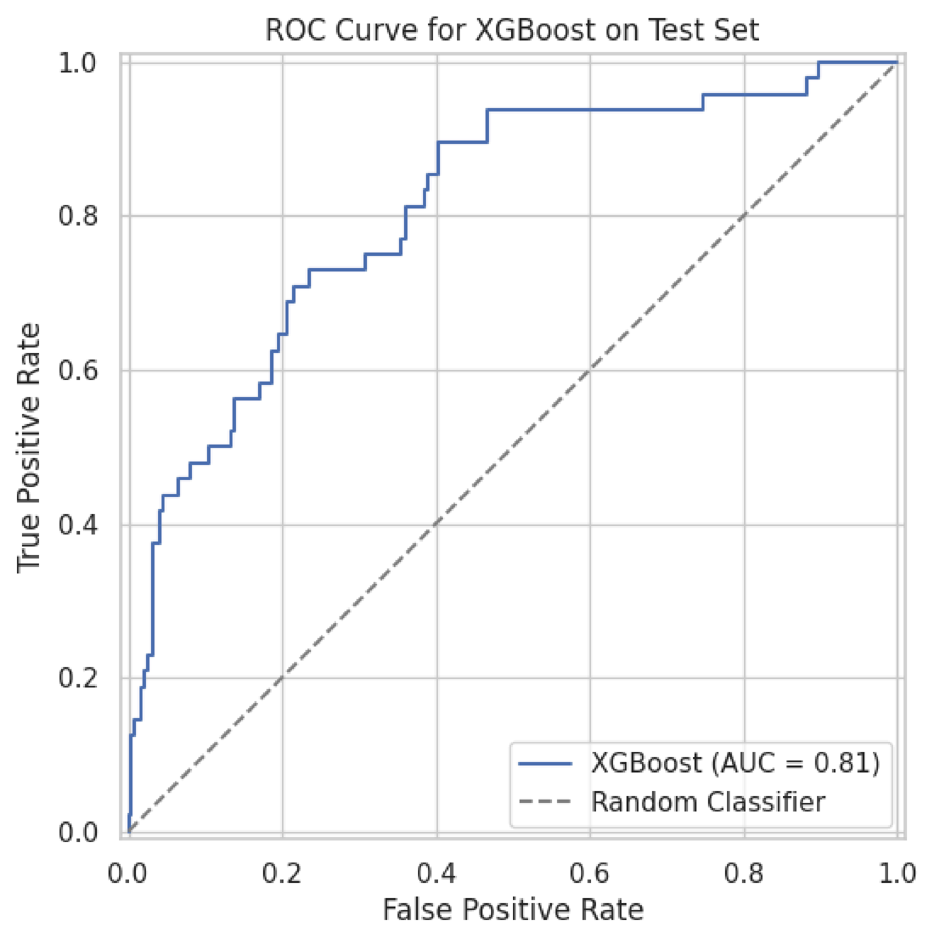
- Preprocessing:
  - Encoded categorical variables (e.g., JobRole, Department, satisfaction levels).
  - Created derived features: age bands, tenure bands, stress gap, welfare index.
  - Train–test split with stratification to preserve attrition ratio.
  - Standard scaling for numeric input to the model.
- Model:
  - Chosen model: XGBoost classifier (widely used in tabular ML).
  - Reason: handles non-linear relationships, interactions, mixed feature types, robust baseline for HR tabular data.
- Evaluation metrics:
  - F1-score (balance precision & recall on the “leave” class).
  - ROC-AUC (overall discriminative power).
  - Confusion matrix to understand errors.

MODEL PERFORMANCE – IS THE POC USEFUL?

The model achieves reasonable discriminative power, making it useful for prioritizing high-risk employees, not for perfect prediction.

- Achieved good ROC-AUC (better than random, captures signal in data).
- F1-score on the "leave" class shows that the model can identify a substantial portion of leavers at acceptable error levels.
- Confusion matrix shows:
  - Some false positives (flagged at risk but stay) → acceptable for HR follow-up.
  - Some false negatives (missed leavers) → opportunity for future improvement (more features, more data).

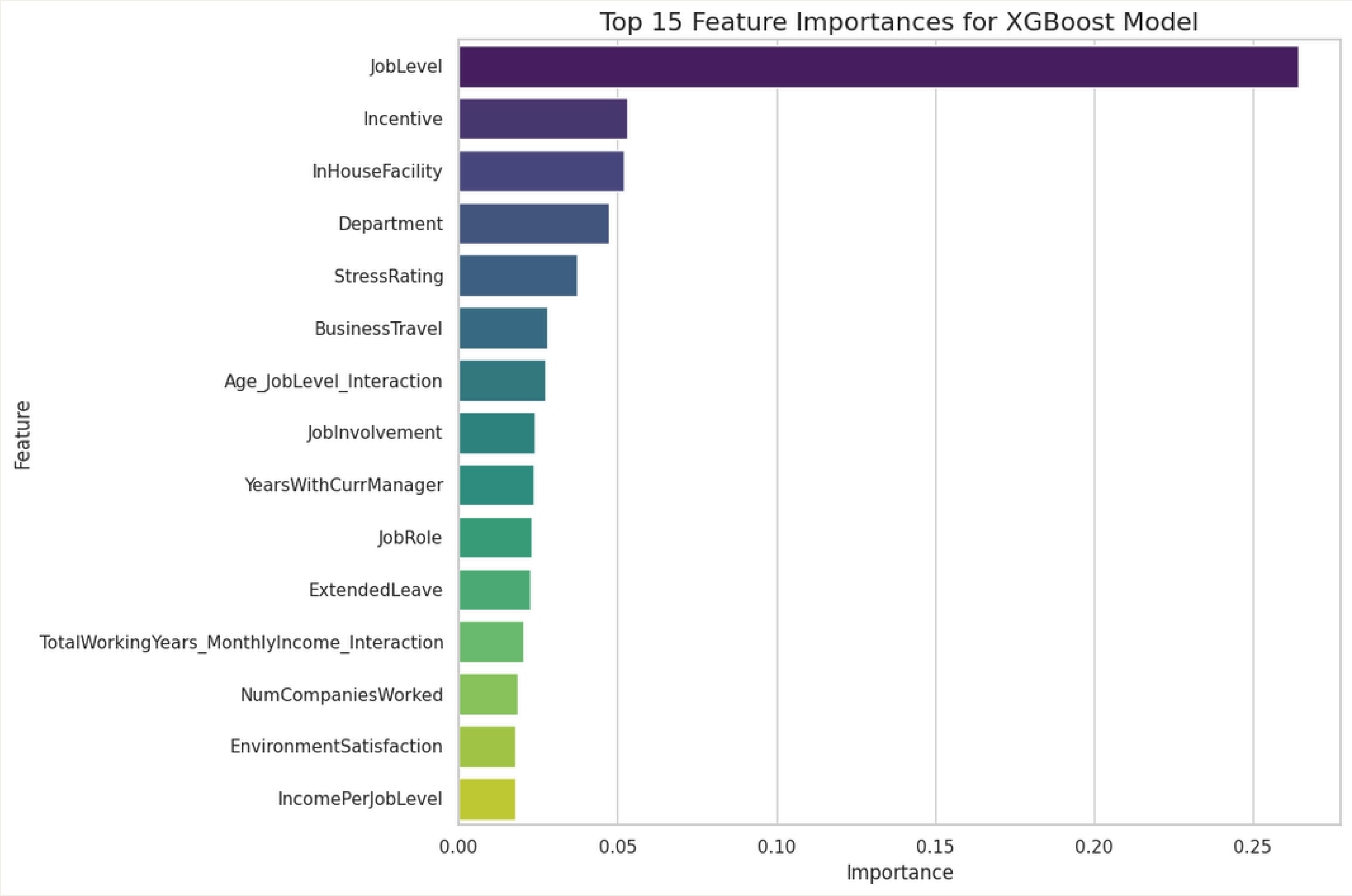




WHAT DRIVES ATTRITION? (FEATURE IMPORTANCE)

Stress, incentives, work style, and engagement features are among the top drivers of attrition.

- Top features from XGBoost importance:
  - JobLevel
  - Incentive / MonthlyIncome
  - InHouseFacility
  - Department
  - Stress Rating
  - Business Travel
- Interpretation:
  - Employees facing high stress, limited career progression (job level), lower incentives, and frequent travel are at the greatest risk of leaving.
  - Meanwhile, access to supportive facilities and placement within healthier departments significantly improves retention.



BUSINESS RECOMMENDATIONS

Data-Driven Retention Strategy for IBM

- Reduce harmful stress & burnout
  - Identify teams with high StressRating.
  - Introduce workload rebalancing, realistic deadlines, manager coaching, mental health support.
- Expand flexible & remote work options
  - Prioritize flexible/remote options for high-risk roles/segments where feasible.
  - Combine with clear performance expectations.
- Align incentives & perceived fairness
  - Review incentive structures for frontline / high-attrition roles.
  - Introduce small but visible performance-based bonuses or recognition rewards.
- Increase welfare & development usage
  - Promote welfare programs, in-house/external facilities, and extended leave usage.
  - Invest in clear career paths and skill development programs for early-career employees.

QUANTIFIED IMPACT (HIGH-LEVEL)

Potential Business Impact-Even a modest reduction in attrition can generate significant savings.

- Based on industry examples:
  - Replacing a tech employee can cost ~50–60% of annual salary (recruiting, onboarding, ramp-up).
  - For a mid-sized consulting workforce, even a small % drop in attrition can save hundreds of thousands to millions per year.
- If IBM has 1,500 employees and reduces attrition by just a few percentage points, it avoids dozens of exits and their associated costs.
- The ML model allows IBM to:
  - Prioritize high-risk employees for interventions.
  - Monitor if retention initiatives (flexibility, welfare, stress programs) are working over time.

IMPLEMENTATION ROADMAP

The PoC can be scaled into a live HR analytics tool with clear steps.

- Short term (PoC → Pilot)
  - Integrate model with HR data pipeline (regular refresh).
  - Dashboard for HR: risk scores, top drivers per employee segment.
  - Run small-scale pilots of retention initiatives in high-risk teams.
- Medium term
  - Gather feedback from HR and managers.
  - Fine-tune model (additional features: engagement surveys, performance history, project data).
  - Track before/after attrition in pilot groups.
- Long term
  - Institutionalize data-driven retention as part of HR strategy.
  - Combine with survey analytics & qualitative insights (exit interviews).

REFERENCES & DATA SOURCES

References & Data Sources

- Dataset: IBM HR Analytics Employee Attrition & Performance (modified for course).
- Industry / literature:
  - FutureCode (2025) – All About Employee Attrition in IT and How It Differs from Employee Turnover. https://future-code.dev/en/blog/all-about-employee-attrition-in-it-and-how-it-differs-from-employee-turnover
  - Bucketlist Rewards (2025) – The True Cost of Employee Turnover in Tech. https://bucketlistrewards.com/blog/the-true-cost-of-employee-turnover-in-tech/
  - Pallathadka et al. (2022) – Attrition in software companies: Reason and measures (Materials Today: Proceedings). https://www.sciencedirect.com/science/article/pii/S2214785321042231
- Tools & libraries:
  - Python, pandas, scikit-learn, XGBoost, matplotlib / seaborn.